

Income-Based Modeling for Apartment Building: Predicting Gross Income Multiplier

Project 3 | BUSI 444

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Introduction

This report examines the income-based modeling of apartment building sales, employing a similar methodology to Lesson 7's framework. The primary objective is to construct a robust model for estimating the gross income multiplier (GIM), using 30 sales for the apartments. Initially, a coefficient of variation (COV) analysis was conducted on SP_UNIT, SP_SQFT, and GIM, revealing GIM as the most suitable dependent variable due to its lower COV. The subsequent analysis of variables, including transformations and correlation assessments, resulted in a final model derived through stepwise regression. This model incorporates key factors such as the presence of an elevator, neighborhood, number of stories, bachelor units, small unit indicator, and effective age. The model's efficacy was validated through ratio statistics and Kruskal-Wallis tests, demonstrating its accuracy in estimating GIM, effective income, and price, with coefficient of variation (COV) values of 6.61%, 6.36%, and 6.61% for PRICE_RATIO, EFF_INC_RATIO, and PRICE_RATIO, respectively. The results of the Kruskal-Wallis test indicate that the model effectively captures variations in GIM, effective income, and price across different property segments.

Choose a Valuation Model

The dataset for this project is the same as the dataset for lesson 7 and so I skip describing the variables in the dataset. Following the general framework in lesson 7, I am using an additive model in this section. First, I need to find the response variable for the model. Referring to the video for project 3, I am doing COV analysis on SP_UNIT (sale price per unit), SP_SQFT (sale price per squared feet), and GIM (gross income model) to pick the right dependent variable from those.

Coefficient of Variable for Three Dependent Variable

	SP_UNIT	SQ_SQF	GIM
COV	0.212	0.265	0.200

Based on the table above, I am proceeding with GIM as the dependent variable. Following the model provided in Lesson 7, I am proposing the model below:

$$\text{GIM} = b_0 + (b_1 \times \text{LOCATION}) + (b_2 \times \text{AGE}) + (b_3 \times \text{UNITS}) + (b_4 \times \text{SUITSIZE}) \\ + (b_5 \times \text{LANDRATIO}) + (b_6 \times \text{AMENITIES})$$

Having established GIM as the response variable, the next step involves a thorough review and examination of the dataset's variables to understand their potential relevance to the model.

Review and Examine the Variables

Since this dataset is the same as in Lesson 7, this section replicates the corresponding analysis. Here, I am providing a summary of the variable review. The sales data covers January 31, 2025, to February 27, 2017, for apartments with 4 to 120 units, but more than 50% of them have fewer than 20 units. There is an outlier in building area with 135,147 square feet, but more than 50% of the cases have less than 15,400 square feet. Similarly, there is an outlier in land size with 224,334 square feet, but around 60% of the data has lot sizes under 27,000 square feet. Apartments are between 25 to 50 years old, but around 80% were built between 1977 and 1992, with a large number constructed in 1991 alone.

With an understanding of the variables' distributions and ranges, the next step involves applying appropriate transformations to prepare the data for modeling.

Transformations

Similar transformations as those explained in Lesson 7 are followed here too. I am creating a new column SML_UNIT that is 1 if the apartment has 18 or fewer and 0 otherwise. Other transformations are listed in Appendix (Table A.1), including transformations for neighborhoods, conditions and stories. Following the instruction in the Video for Project 3, I am only including the Syntax for variable transformation in the Appendix and not for all the steps. As I already spoke with my tutor, I am using R as my programming language. Having transformed the variables as necessary, the analysis proceeds to explore the relationships between these variables and the response variable, GIM, through scatterplots and boxplots.

Relationship with Variables

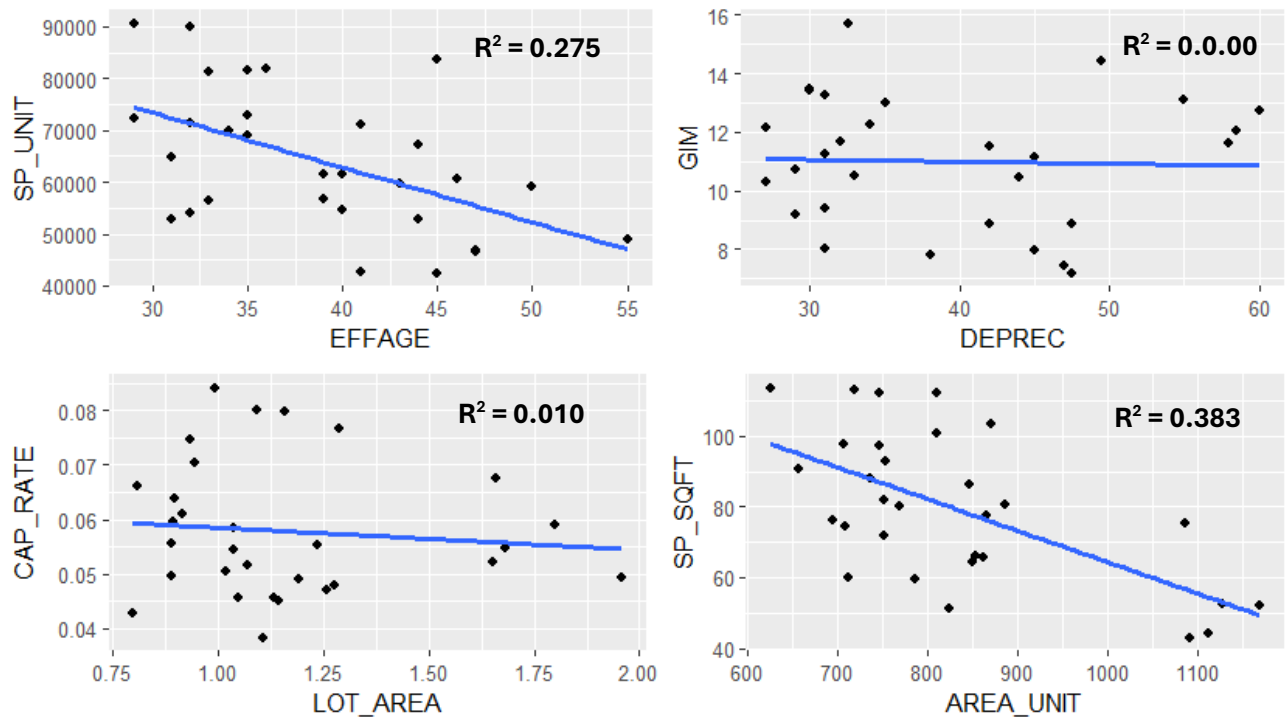
Scatterplot

To understand the relationship between the variables, I first consider the Pearson correlation coefficients between our return variables (SP_UNIT, SP_SQFT, CAP_RATE, and GIM) and continuous variables (EFFAGE, DEPREC, LOT_AREA, AREA_UNIT). High correlation coefficient indicates higher influence of the measure of return. Not surprisingly, I come up with the same results as those in Lesson 7.

Correlation between Variables

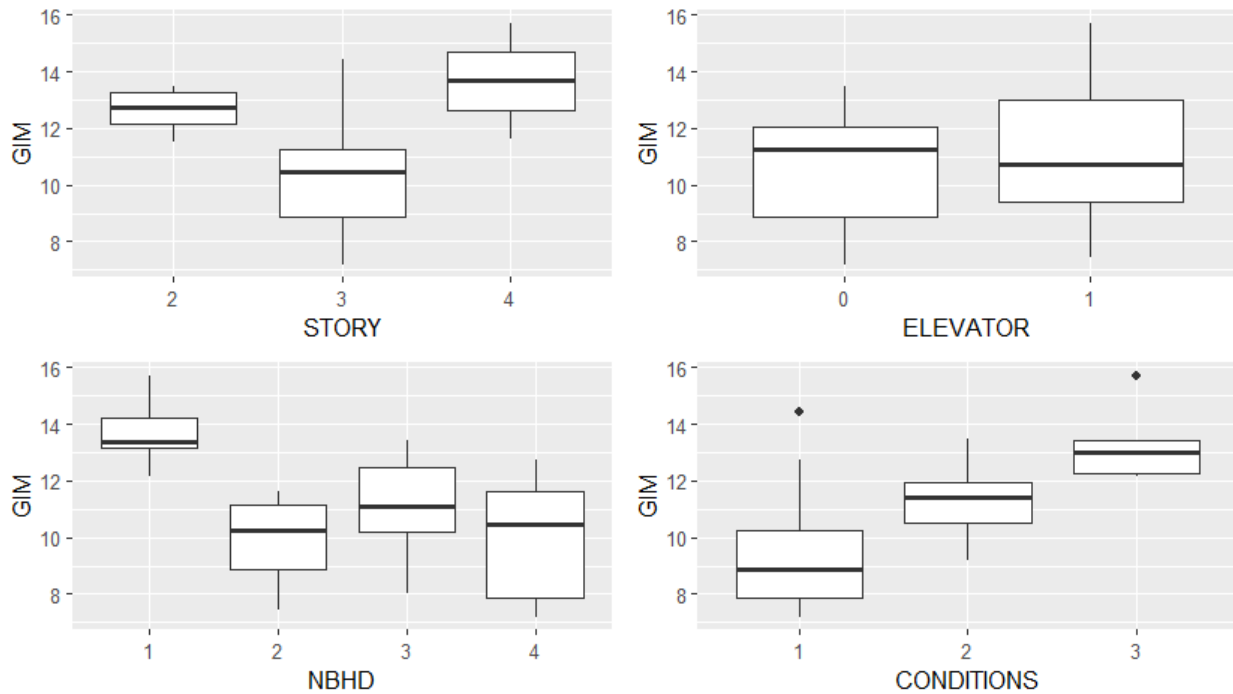
	EFFAGE	DEPREC	LOT_AREA	AREA_UNIT	EXP_EGI
SP_UNIT	-0.525	-0.040	0.085	-0.091	-0.230
SP_SQFT	-0.600	-0.169	0.153	-0.619	-0.174
CAP_RATE	0.744	0.109	-0.101	0.377	0.051
GIM	-0.692	-0.031	0.098	-0.280	-0.141

Some of the scatter plots corresponding the same variables shown in the table above are considered below and more in Appendix (Fig. A.1 – A.4). Note that the value for R^2 is always the squared of the Pearson correlation coefficient given in the table above. The table above and the scatter plots show that EFFAGE and AREA_UNIT have high correlation with the response variable and should be included in the model. In addition to examining continuous variables, the relationship between categorical variables and GIM is explored using boxplots to provide further insights into potential model predictors.



Boxplots

Similarly, the relationship between the categorical variables and the response variable is shown using box plots. Here, I include only the box plots for the selected response variable (GIM). Four box plots are included in the main text, while the rest can be found in the Appendix (Fig. A.5). Following Lesson 7, I do not consider unit-specific variables such as ONEBED or TWOBED, as we are evaluating building market value, and their effects should be captured by other variables.



The results show that buildings in better condition tend to have higher GIM, which is expected-investing in well-maintained buildings leads to faster returns. Apartments with three stories appear to have the lowest GIM compared to those with two or four stories. Having an elevator slightly increases GIM, and neighborhood 1 has the highest GIM compared to other neighborhoods. Additionally, buildings with fewer units tend to have higher GIM and faster returns, supporting the inclusion of the new column in the model. A lower number of bachelor units also increases GIM, as do factors like under-parking and heat type 2, though the effect of the latter is not significant. To further refine the variable selection process and identify potential multicollinearity issues, a correlation matrix is examined

Correlation Matrix

The correlation matrix Table is included in the Appendix and it shows the correlation coefficients between different variables. Those with high correlation (above 0.5 or less than -0.5) are highlighted with red indicating they could cause collinearity in the model. According to the table, EFFAGE and CONDITION_1 and CONDITION_3 have high correlation as well as SML_UNITS and UNITS. Since these variables are representing similar things, we should include only one of them in the model. Not surprisingly, the results here are consistent with finding in Lesson 7 as the independent variables are the same.

Given the plots and the table in this section, I am finalizing the list of variables as follows:

Either: UNITS or SML_UNIT
 Either: EFFAGE or CONDITION_1 and CONDITONS_3
 Common: UND_PARK
 ELEVATOR
 LOT_AREA
 AREA_UNIT
 NBHD
 STORY
 HEATTYPE
 BACH

Note that the last two common variables are not listed in Lesson 7 but I am including them as well as they have been shown to have high correlation with the response variable GIM. With a refined set of variables, the next step involves variable selection and calibration to build the most effective model.

Variable Selection and Calibration

Following a similar discussion in Lesson 7, I am introducing the following variable groups

Group 1	Common + UNITS + EFFAGE
Group 2	Common + SML_UNIT + EFFAGE
Group 3	Common + SML_UNIT + EFFAGE
Group 4	Common + UNITS + CONIDTIONS

Table below shows the statistics for model summary for the models above.

	R ²	Adjusted R ²	SEE	F	Sig.	Max VIF
Model 1	0.915	0.845	0.632	13.176	0.000	3.698
Model 2	0.892	0.792	0.709	8.894	0.000	4.056
Model 3	0.940	0.891	0.529	19.280	0.000	3.697
Model 4	0.872	0.753	0.773	7.305	0.000	5.258

Table above shows Model 3 has the highest adjusted R² as well as highest F statistics. However, all models have a VIF larger than 3.33 which causes potential collinearity. This issue should be addressed for calibrating the variables to ensure that no variable have VIF larger than 3.33. Based on the results, I am following with Model 3, which results in the following coefficients.

ANOVA

	Sum of Squares	df	Mean Square	F	Sig.
Regression	131.66	13	10.13	19.28	0.00
Residuals	8.41	16	0.53		
Total	140.06	29			

Coefficient

Variable	Coefficient	Standard_Coefficient	t_Statistic	Significance	Tolerance	VIF
(Intercept)	13.185	5.999	6.278	0.000		
UND_PARK	0.869	0.395	1.692	0.110	0.271	3.697
ELEVATOR	0.460	0.209	1.002	0.331	0.339	2.953
LOT_AREA	1.205	0.548	1.947	0.069	0.516	1.936
AREA_UNIT	0.001	0.001	1.024	0.321	0.455	2.196
NBHD_1	0.945	0.430	1.729	0.103	0.367	2.728
NBHD_3	-0.797	-0.363	-1.428	0.173	0.287	3.482
NBHD_4	-0.269	-0.122	-0.649	0.525	0.571	1.750
STORY_2	1.241	0.565	2.814	0.012	0.503	1.989
STORY_4	0.957	0.435	1.583	0.133	0.770	1.299
HEATTYPE	-0.519	-0.236	-0.763	0.457	0.607	1.647
BACH	-0.121	-0.055	-1.470	0.161	0.434	2.305
SML_UNIT	1.662	0.756	3.853	0.001	0.377	2.655
EFFAGE	-0.148	-0.067	-4.181	0.001	0.315	3.176

The table of coefficients show that UND_PARK has the $VIF > 3.33$ which causes collinearity. This is expected since UND_PARK shows to have high correlation with EFFAGE and hence high VIF. Since EFFAGE has higher effect on the response variable, I will keep EFFAGE and remove UND_PARK to reduce the VIF. The results after removing UND_PARK are available in the Appendix (Table A.3). The table shows that removing UND_PARK solves the problem with VIF and the VIF for the coefficients in the new model are all below 3.33. With the issue of multicollinearity addressed, the next step is to refine the model by removing statistically insignificant coefficients. This is achieved through stepwise regression.

Stepwise Regression

To refine the model by removing statistically insignificant coefficients, a stepwise regression method was employed. This process led to the removal of NBHD_3, NBHD_4, and HEATTYPE from the model. The detailed steps of the stepwise regression, along with the detailed coefficient values at each step, are included in the Appendix (Table A.4).

Stepwise Method

Model	Predictors:
Model1	ELEVATOR, LOT_AREA, AREA_UNIT, NBHD_1, NBHD_3, NBHD_4, STORY_2, STORY_4, H EATTYPE, BACH, SML_UNIT, EFFAGE
Model2	ELEVATOR, LOT_AREA, AREA_UNIT, NBHD_1, NBHD_3, NBHD_4, STORY_2, STORY_4, B ACH, SML_UNIT, EFFAGE
Model3	ELEVATOR, LOT_AREA, AREA_UNIT, NBHD_1, NBHD_4, STORY_2, STORY_4, BACH, SML _UNIT, EFFAGE
Model4	ELEVATOR, LOT_AREA, AREA_UNIT, NBHD_1, STORY_2, STORY_4, BACH, SML_UNIT, EF FAGE

ANOVA

	Sum of Squares	df	Mean Square	F	Sig.
Regression	128.92	8	16.11	30.36	0.00
Residuals	11.15	21	0.53		
Total	140.06	29			

Coefficients

	Coefficients	Standard_Coefficient	t_Statistic	Significance	Tolerance	VIF
(Intercept)	13.832	6.294	8.915	0.000		
ELEVATOR	0.708	0.322	1.771	0.091	0.451	2.217
LOT_AREA	0.742	0.338	1.384	0.181	0.695	1.438
NBHD_1	1.653	0.752	4.375	0.000	0.775	1.291
STORY_2	0.873	0.397	2.206	0.039	0.631	1.584
STORY_4	1.132	0.515	1.916	0.069	0.815	1.227
BACH	-0.176	-0.080	-2.671	0.014	0.677	1.477
SML_UNIT	1.456	0.663	3.645	0.002	0.443	2.255
EFFAGE	-0.133	-0.061	-4.290	0.000	0.414	2.413

And therefore, the final model is achieved as follows:

$$\text{GIM} = (\text{INTERCEPT}) + 0.708 (\text{ELEVATOR}) + 0.742 (\text{LOT_AERA}) + 1.653 (\text{NBHD_1}) \\ + 0.873 (\text{STORY_2}) + 1.132 (\text{STORY_4}) - 0.176 (\text{BACH}) \\ + 1.456 (\text{SML_UNIT}) - 0.133 (\text{EFFAGE})$$

The model's coefficients generally align with expectations: elevators, increased stories, and smaller units positively correlate with higher GIM, as does property location in neighborhood 1. Conversely, bachelor units and older effective age negatively impact GIM. While a larger lot area exhibits a positive relationship with GIM, its effect is statistically less significant. *It is important to mention that while the LOT_AREA does not seem to be statistically significant, including it results in lower AIC than removing it and hence stepwise process ends up*

including it in the model. The model summary shows no variables with a VIF exceeding 3.33 indicates that multicollinearity is unlikely to be a concern in this model. With the final model established, the next section details the analysis of the results, comparing predicted values against actual values to assess the model's accuracy and effectiveness. Using the model, I am predicting the GIM_PRED which is the predicted value for GIM based on the model above. Next section, I am analysing the results by comparing the predicted values and the actual values.

Analysis of Results

Following the discussion in Lesson 7, I am defining the following variables:

Variable	Description
$GIM_RATIO = GIM_PRED / GIM$	Ratio of predicted GIM to actual GIM
$EFF_INC_PRED = PRICE / GIM_PRED$	Effective Income Prediction using predicted GIM
$EFF_INC_RATIO = EFF_INC_PRED / EFF_INC$	Ratio of predicted effective income to actual effective income
$PRICE_PRED = EFF_INC * GIM_PRED$	Price prediction using predicted GIM and Effective Income
$PRICE_RATIO = PRICE_PRED / PRICE$	Ratio of predicted price to actual price

Following table shows the ratio statistics for the GIM_RATIO and EFF_INC_RATIO

	PRICE_RATIO	EFF_INC_RATIO	PRICE_RATIO
MEAN	1.00	1.00	1.00
MEDIAN	1.00	1.00	1.00
STD.DEV	0.07	0.06	0.07
COV	6.61%	6.36%	6.61%
Percentile	0.93	0.94	0.93
	0.95	0.96	0.95
	1.04	1.05	1.04
	1.06	1.08	1.06

The median and the mean are very close to one for both ratios. Also, the standard deviation is small relative to the mean. In fact, the COVs are 6.61%, 6.36%, and 6.61% for PRICE_RATIO, EFF_INC_RATIO, and PRICE_RATIO respectively. Also, the percentiles are close enough to one for all ratios showing the accuracy of the model. The results show that the model is not only good for estimating GIM, but also EFF_INC and PRICE. Suggested by

Lesson 7, I finalized this section by doing K-W test for different variable to ensure the prediction is equal and fair. The table below summarizes the K-W test for different variables.

Title	df	p.value	CH
NBHD	3	0.800	1.005
STORY	2	0.984	0.033
CONDITN	2	0.860	0.302
SML_UNIT	1	0.917	0.011
ELEVATOR	1	0.950	0.004
BACH	6	0.822	2.891

Interestingly, the p-value for all the tests is well above 5%, indicating that the evaluation is equitable across different variables. Consequently, no further adjustments are needed, and the final model remains as derived previously:

$$\begin{aligned} \text{GIM} = & (13.832) + 0.708 (\text{ELEVATOR}) + 0.742 (\text{LOT_AERA}) + 1.653 (\text{NBHD_1}) \\ & + 0.873 (\text{STORY_2}) + 1.132 (\text{STORY_4}) - 0.176 (\text{BACH}) \\ & + 1.456 (\text{SML_UNIT}) - 0.133 (\text{EFFAGE}) \end{aligned}$$

Conclusion

This study successfully developed and validated an income-based model for estimating apartment building Gross Income Multiplier (GIM) using 30 apartment sales. GIM is selected as the dependent variable due to its low Coefficient of Variation (COV) of 0.2. Stepwise regression refined the model, confirming that factors like elevators, specific neighborhoods, number of stories, and smaller unit indicator positively correlate with GIM, while bachelor units and effective age have a negative impact. Multicollinearity was addressed, ensuring all Variance Inflation Factors (VIFs) remained below the 3.33 threshold. The model demonstrated strong predictive accuracy, with ratio analyses yielding Mean and Median values near 1.00 and low COVs for price, effective income, and GIM ratios (6.61%, 6.36%, and 6.61% respectively). Furthermore, Kruskal-Wallis tests confirmed equitable predictions across different property variables, with all p-values exceeding the 5% significance level, validating the model's robustness for estimating GIM.

Appendix

Transformations

I am doing the following transformations:

Table A 1. Transformation of Variables

Variable	Description
$SML_UNIT = 1 \text{ IF } UNIT \leq 18, 0 \text{ otherwise}$	Indicator if the apartment has 18 or fewer units
$SP_UNIT = PRICE/UNIT$	Sale price per unit
$SP_SQFT = PRICE/AREA$	Sale price per square foot
$CAP_RATE = NOI/PRICE$	Capitalization rate
$GIM = PRICE/EFF_INC$	Gross income multiplier
$LOT_AREA = LOTSIZE/AREA$	Ratio of lot size to building area
$AREA_UNIT = AREA/UNIT$	Building area per unit
$EXP_EGI = EXPENSES/EFF_INC$	Expense ratio
$GI_UNIT = GROSS_IN/UNITS$	Gross income per unit
$NOI_UNIT = NOI/UNITS$	Net operating income per unit
$NBHD_1 = 1 \text{ IF } NBHD = 1, 0 \text{ otherwise}$	Indicator if the property is in NBHD = 1
$NBHD_3 = 1 \text{ IF } NBHD = 3, 0 \text{ otherwise}$	Indicator if the property is in NBHD = 3
$NBHD_4 = 1 \text{ IF } NBHD = 4, 0 \text{ otherwise}$	Indicator if the property is in NBHD = 4
$CONDITION_1 = 1 \text{ IF } CONDITION = 1, 0 \text{ otherwise}$	Indicator if the property's conditions is 1
$CONDITION_3 = 1 \text{ IF } CONDITION = 3, 0 \text{ otherwise}$	Indicator if the property's conditions is 3
$STORY_2 = 1 \text{ IF } STORY = 2, 0 \text{ otherwise}$	Indicator if the property has 2 stories
$STORY_4 = 1 \text{ IF } STORY = 3, 0 \text{ otherwise}$	Indicator if the property has 4 stories

Relationship between Variables

Scatter Plot

Fig A 1. Scatter Plots Demonstrating the Relationship between Variables and SP_SQFT

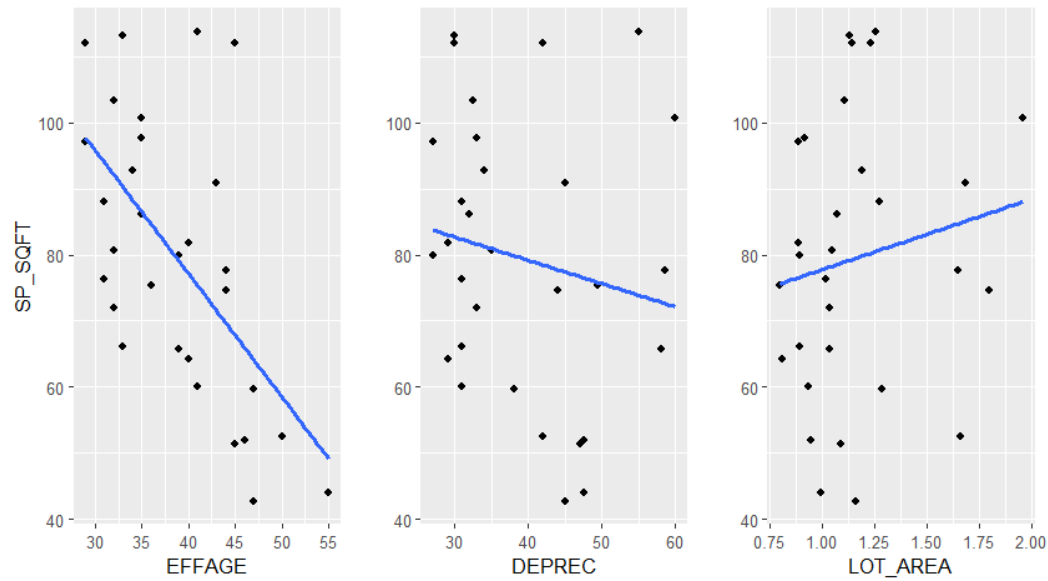


Fig A 2. Scatter Plots Demonstrating the Relationship between Variables and CAP_RATE

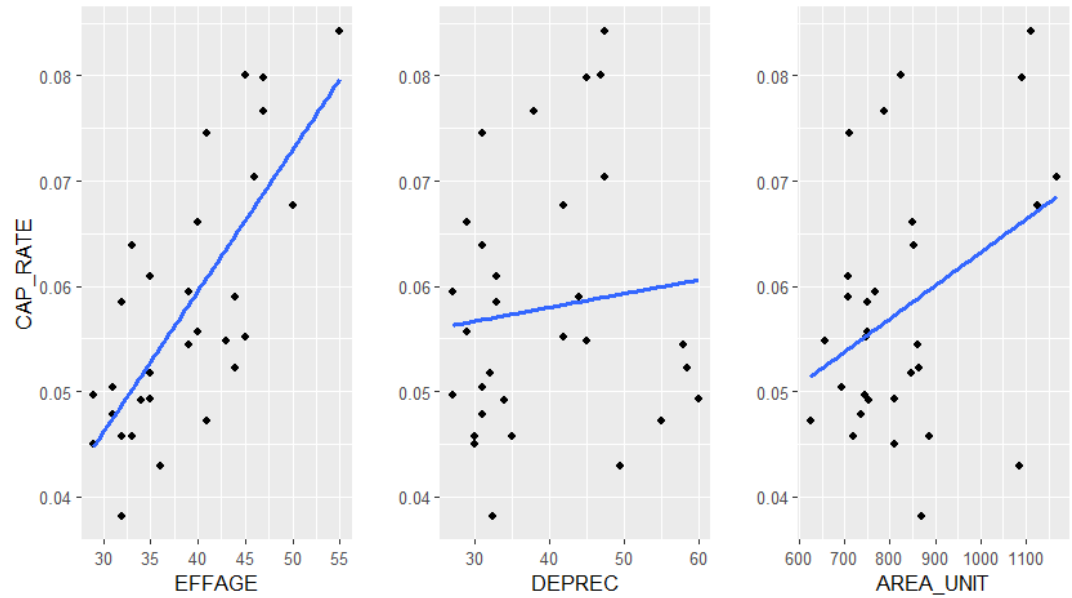


Fig A 3. Scatter Plots Demonstrating the Relationship between Variables and GIM

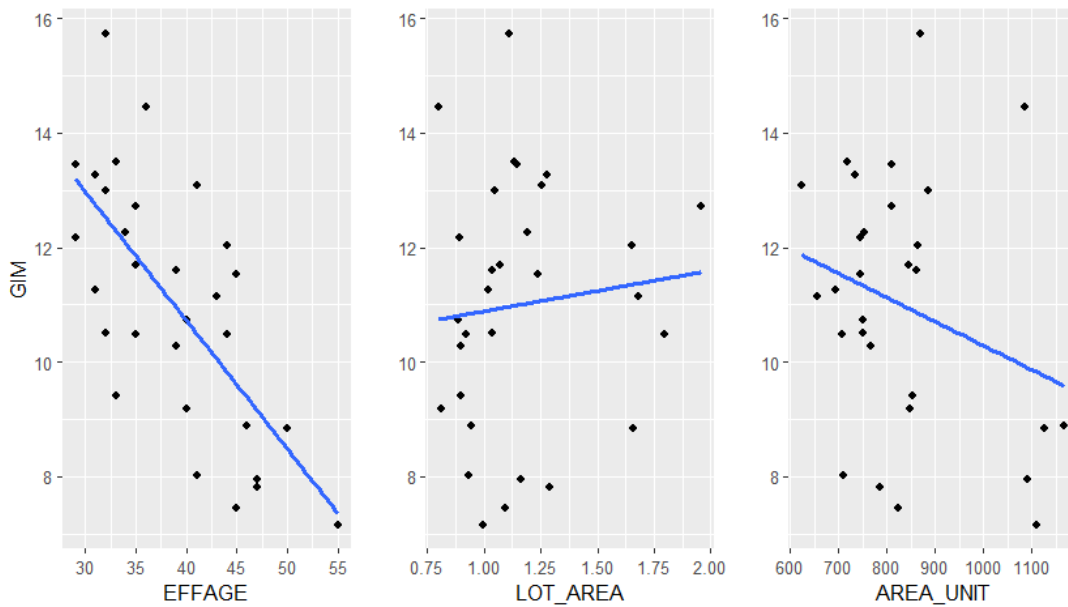
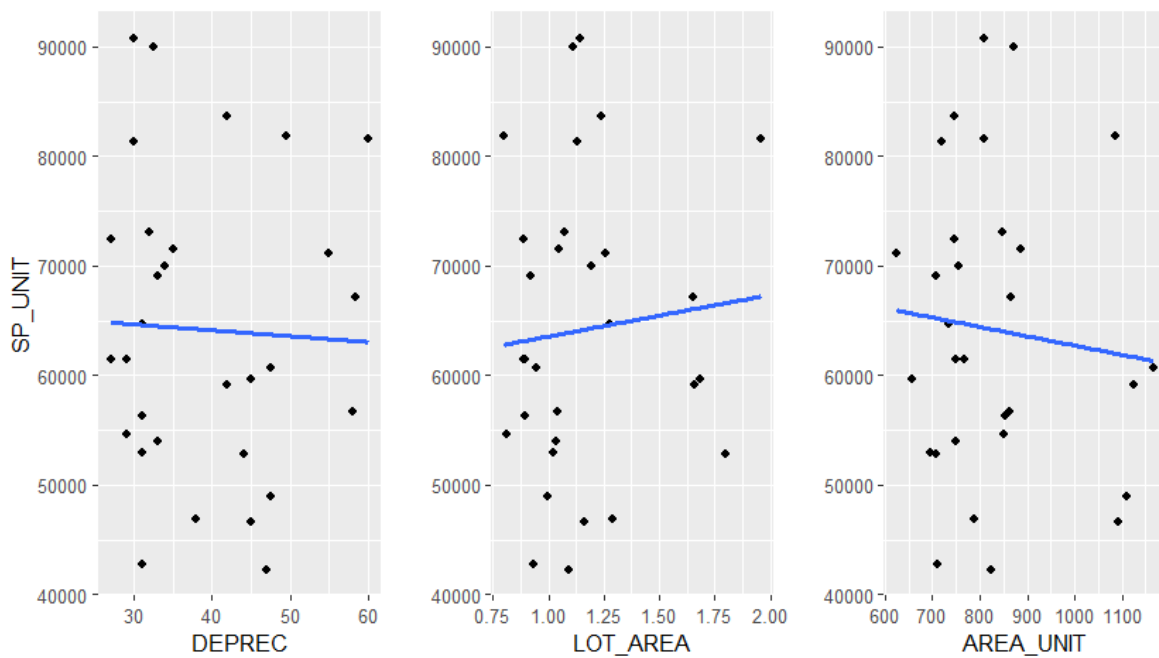
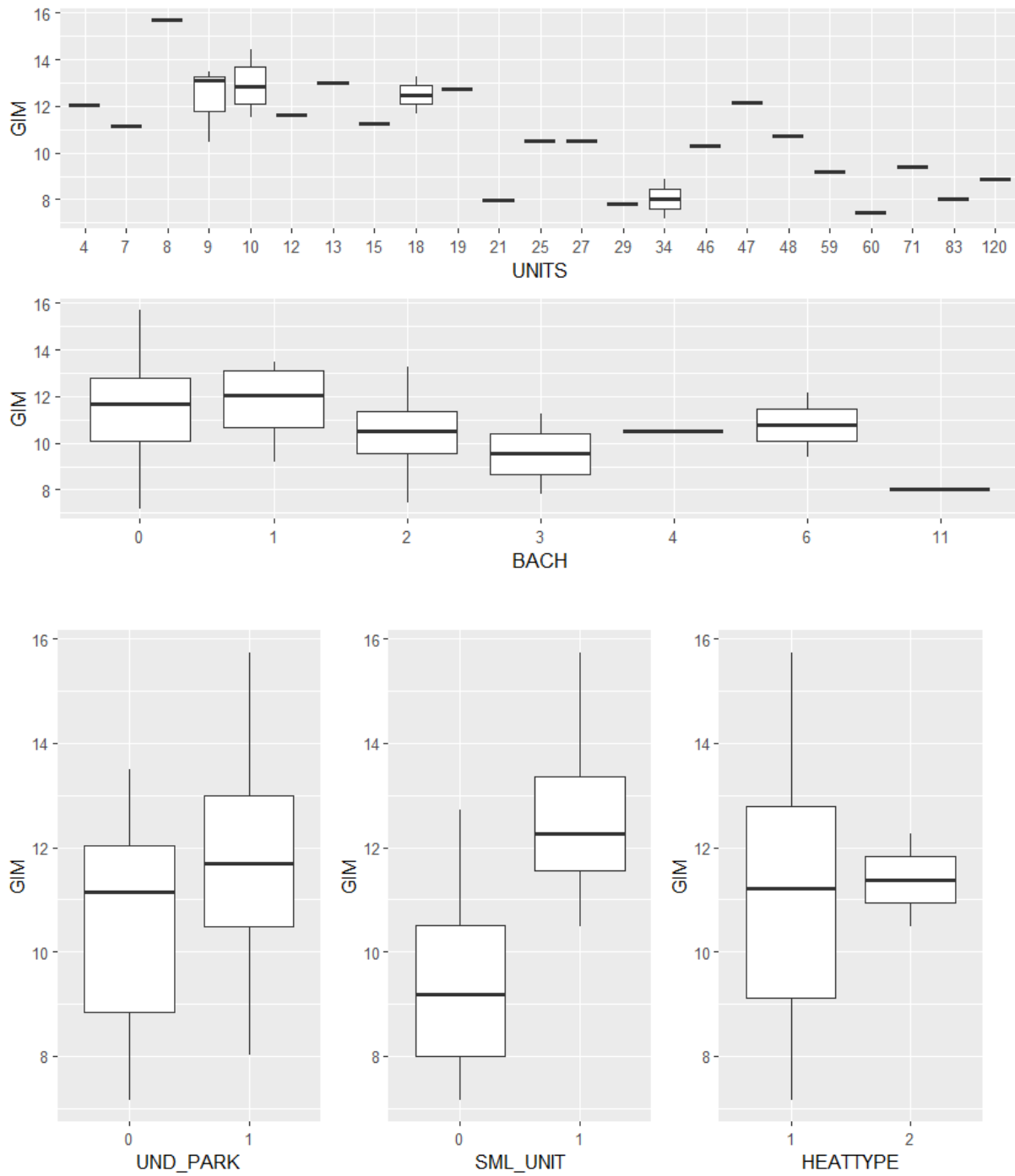


Fig A 4. Scatter Plots Demonstrating the Relationship between Variables and SP_UNIT



Box plots

Fig A 5. Box Plots Demonstrating the relationship between GIM and categorical variables



Variable Selection and Calibration

Table A 3. Model Coefficients and Summary after Removing UN_PARK variable

ANOVA (model 3 after removing UND_PARK)						
	Sum of Squares	df	Mean Square	F	Sig.	
Regression	130.15	12	10.85	18.61	0.00	
Residuals	9.91	17	0.58			
Total	140.06	29				

Coefficient						
Variable	Coefficient	Standard Coefficient	t_Statistic	Significance	Tolerance	VIF
(Intercept)	13.132	5.975	5.936	0.000		
ELEVATOR	0.749	0.341	1.670	0.113	0.393	2.543
LOT_AREA	1.025	0.466	1.596	0.129	0.532	1.879
AREA_UNIT	0.002	0.001	1.125	0.276	0.459	2.177
NBHD_1	1.261	0.574	2.331	0.032	0.415	2.409
NBHD_3	-0.449	-0.204	-0.821	0.423	0.332	3.009
NBHD_4	-0.387	-0.176	-0.900	0.381	0.588	1.701
STORY_2	1.009	0.459	2.285	0.035	0.557	1.796
STORY_4	0.946	0.430	1.485	0.156	0.770	1.299
HEATTYPE	-0.101	-0.046	-0.151	0.882	0.700	1.430
BACH	-0.113	-0.052	-1.310	0.207	0.435	2.298
SML_UNIT	1.661	0.756	3.657	0.002	0.377	2.655
EFFAGE	-0.154	-0.070	-4.151	0.001	0.318	3.144

Stepwise Regression

Coefficients for the Stepwise Regression Models

Table A 4. Details of Coefficients Value during the Stepwise Method.

Model 1				
Term	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	13.132	2.212	5.936	0.000
ELEVATOR	0.749	0.449	1.670	0.113
LOT_AREA	1.025	0.642	1.596	0.129
AREA_UNIT	0.002	0.001	1.125	0.276
NBHD_1	1.261	0.541	2.331	0.032
NBHD_3	-0.449	0.547	-0.821	0.423
NBHD_4	-0.387	0.430	-0.900	0.381
STORY_2	1.009	0.442	2.285	0.035

STORY_4	0.946	0.637	1.485	0.156
HEATTYPE	-0.101	0.668	-0.151	0.882
BACH	-0.113	0.086	-1.310	0.207
SML_UNIT	1.661	0.454	3.657	0.002
EFFAGE	-0.154	0.037	-4.151	0.001
Model 2				
Term	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	12.996	1.965	6.613	0.000
ELEVATOR	0.768	0.419	1.833	0.083
LOT_AREA	1.010	0.618	1.636	0.119
AREA_UNIT	0.002	0.001	1.226	0.236
NBHD_1	1.254	0.524	2.393	0.028
NBHD_3	-0.474	0.507	-0.934	0.363
NBHD_4	-0.396	0.414	-0.957	0.351
STORY_2	1.021	0.423	2.415	0.027
STORY_4	0.951	0.619	1.537	0.142
BACH	-0.111	0.083	-1.341	0.197
SML_UNIT	1.661	0.442	3.760	0.001
EFFAGE	-0.154	0.036	-4.272	0.000
Model 3				
Term	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	12.635	1.920	6.580	0.000
ELEVATOR	0.696	0.410	1.696	0.106
LOT_AREA	1.047	0.614	1.705	0.105
AREA_UNIT	0.001	0.001	1.003	0.328
NBHD_1	1.581	0.389	4.067	0.001
NBHD_4	-0.268	0.389	-0.690	0.499
STORY_2	0.939	0.412	2.278	0.034
STORY_4	1.073	0.603	1.780	0.091
BACH	-0.138	0.077	-1.795	0.089
SML_UNIT	1.544	0.422	3.656	0.002
EFFAGE	-0.140	0.033	-4.291	0.000
Model 4				
Term	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	12.838	1.872	6.856	0.000
ELEVATOR	0.733	0.401	1.827	0.083
LOT_AREA	0.879	0.556	1.580	0.130
AREA_UNIT	0.001	0.001	0.952	0.352
NBHD_1	1.614	0.381	4.236	0.000
STORY_2	0.955	0.406	2.353	0.029
STORY_4	1.106	0.593	1.866	0.077

BACH	-0.141	0.076	-1.856	0.078
SML_UNIT	1.563	0.416	3.759	0.001
EFFAGE	-0.141	0.032	-4.382	0.000
Model 5				
Term	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	13.832	1.551	8.915	0.000
ELEVATOR	0.708	0.400	1.771	0.091
LOT_AREA	0.742	0.536	1.384	0.181
NBHD_1	1.653	0.378	4.375	0.000
STORY_2	0.873	0.396	2.206	0.039
STORY_4	1.132	0.591	1.916	0.069
BACH	-0.176	0.066	-2.671	0.014
SML_UNIT	1.456	0.400	3.645	0.002
EFFAGE	-0.133	0.031	-4.290	0.000

Syntax for Variable Transformation

```
data%<>%mutate(
  SML_UNIT = ifelse(units<=18 , 1, 0),
  SP_UNIT = price/units,
  SP_SQFT = price/area,
  CAP_RATE = noi/price,
  GIM = price/eff_inc,
  LOT_AREA = lotsize/area,
  AREA_UNIT = area/units,
  EXP_EGI = expenses/eff_inc,
  GI_UNI = gross_in/units,
  NO_UNIT = noi/units,
  NBHD_1 = ifelse(nbhd == 1 , 1, 0),
  NBHD_3 = ifelse(nbhd == 3 , 1, 0),
  NBHD_4 = ifelse(nbhd == 4 , 1, 0),
  CONDITION_1 = ifelse(conditn == 1 , 1, 0),
  CONDITION_3 = ifelse(conditn == 3 , 1, 0),
  STORY_2 = ifelse(story == 2 , 1, 0),
  STORY_4 = ifelse(story == 4 , 1, 0))
```